Straddling windows see two levels up the RHM tree

a 5x5 window containing only 2x2 blocks still rejects 90% of grids that are coherent through level 2 -- it is policing the root

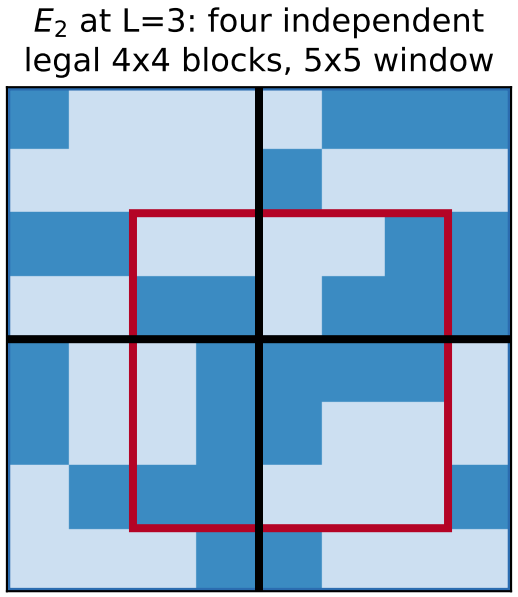
A k x k window contains aligned blocks up to j* = floor(log_s((k+1)/2)).
The question: does it check anything above j*?

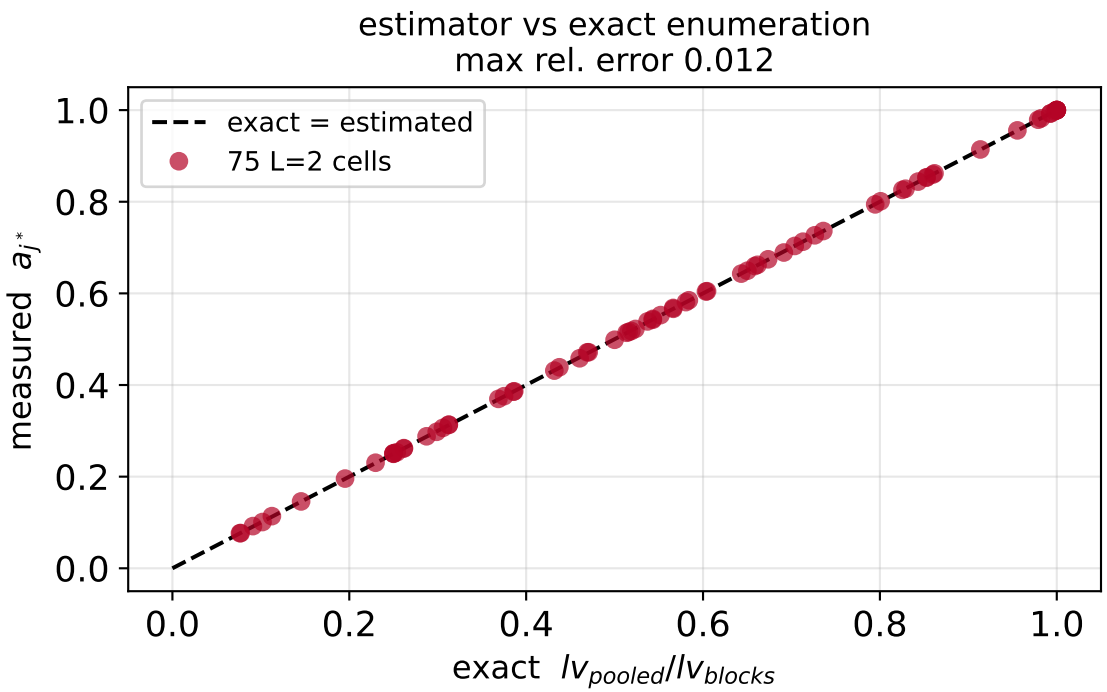
Method: tile a grid with independent legal s^j blocks (ensemble E_j) and measure a_j, how often all its windows are still ones the machine knows.
a_L = 1 verifies the window set is complete; a_j < 1 for j > j* is the signal.

Validated against exact enumeration of every L=2 grid, then run at L=3 where enumeration is impossible (v^64 grids).

160 cells, 146 with a provably saturated window set.

1. The measurement, and its exact check



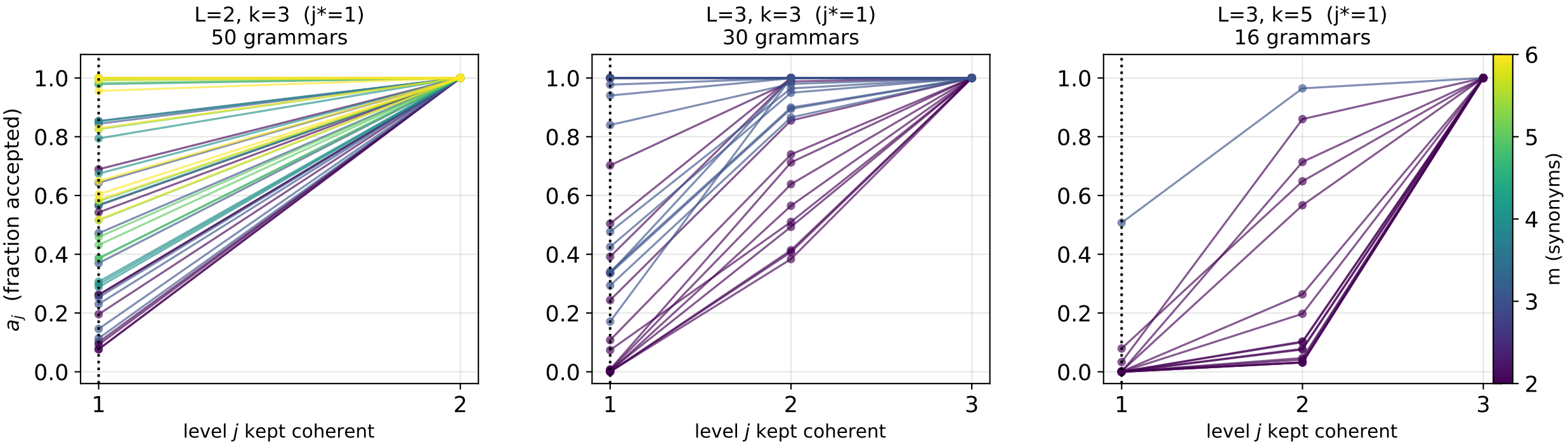


A k x k window fully contains an aligned block of side b only when k >= 2b-1, so it directly certifies levels up to j* = floor(log_s((k+1)/2)). The question is whether it also polices the levels ABOVE j*, using windows that straddle a block boundary and see the right edge of one block against the left edge of the next.

Ensemble E_j tiles the grid with independent uniform draws from R_j, the distinct renderings of an aligned s^j block: coherent through level j, scrambled above it. a_j is the fraction of E_j grids whose every window lies in the machine's window set W. a_L = 1 says W is saturated; a_j < 1 for j > j* says the window sees above j*.

At L=2 every grid can be enumerated, so a_{j*} has a closed-form check -- the right panel. L=3 has v^64 grids and only the estimator reaches it.

2. The window polices two levels above the block it contains

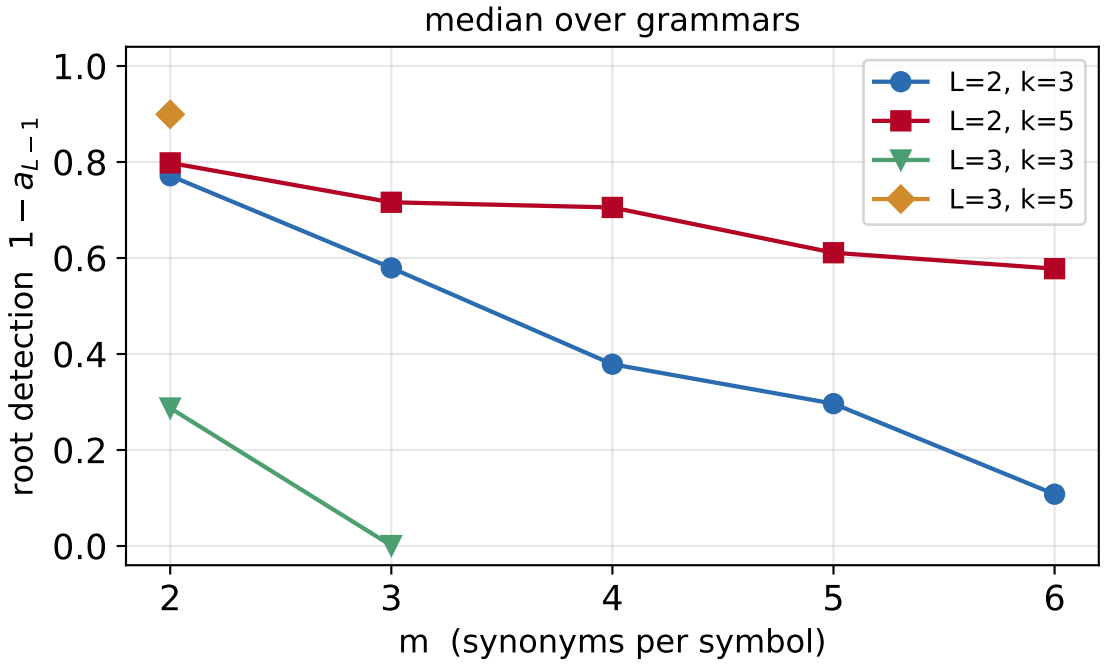


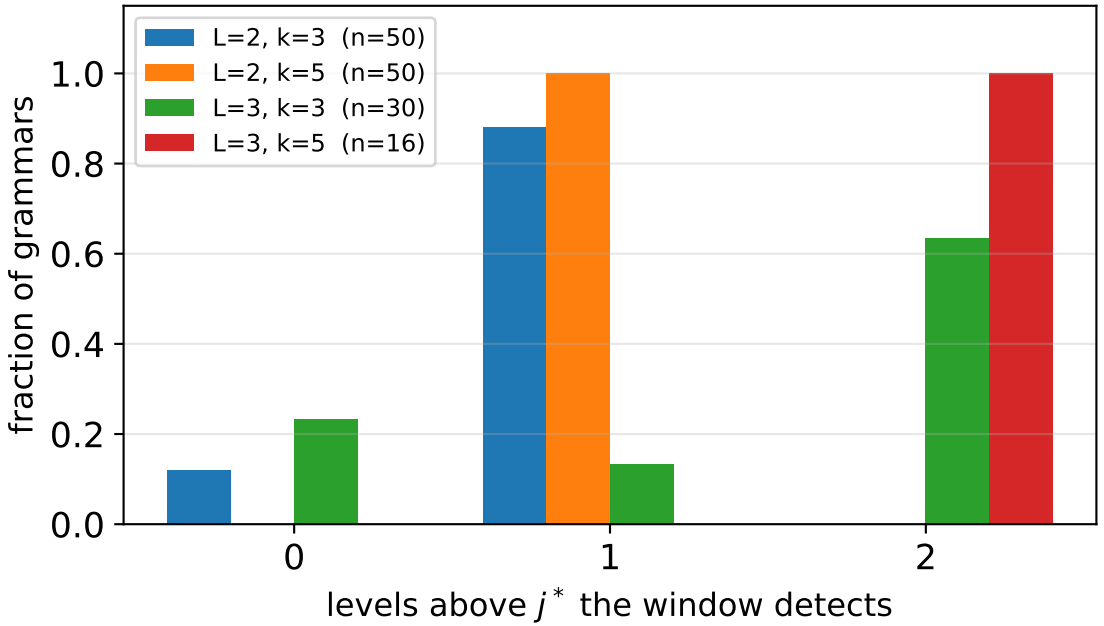
Each line is one grammar; the dotted line is j*, the deepest level the window contains outright. a_j = 1 at j = L is the saturation check and holds everywhere plotted. The interesting point is j = j*+1 and beyond: a value below 1 there means the window rejects grids that are ALREADY coherent at that level, so its reach is strictly deeper than j*.

At L=3, k=5 (j* = 1) the window contains only 2x2 blocks, yet 16 of 16 grammars have a_2 < 0.99 -- median a_2 = 0.10. Those rejections can only come from windows straddling the boundary between two 4x4 level-2 blocks, i.e. the window is checking the ROOT, two levels above j*. That is the effect hshunt predicted.

a_1 is far smaller still (median 0.000): scrambling at level 1 breaks far more windows than scrambling only the root does.

3. The signal dies as the grammar above j* gets richer





Left: the signal is set by how constrained a boundary between two sibling blocks is, not by the window size alone. Every extra synonym adds legal boundary pairs, so fewer scrambled pairings are detectable and root detection falls monotonically in m -- from 0.77 to 0.11 across m=2..6 at L=2, k=3. A wider window (k=5) decays far more slowly.

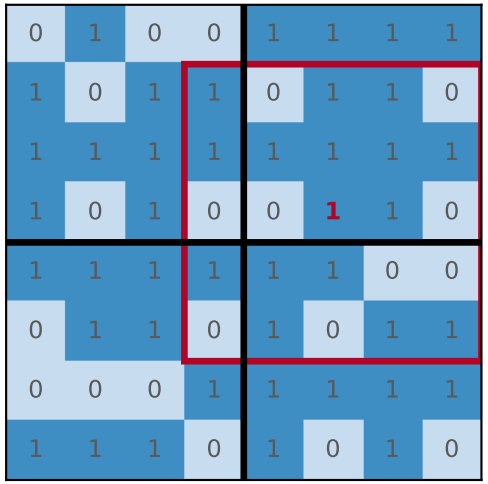
Right: how far the reach actually extends. At L=3 with j* = 1 a window could in principle see one level up (level 2) or two (the root). All 16 of 16 saturated k=5 grammars see two; at k=3 it is 19 of 30, with 7 seeing nothing at all. So the reach is not fixed by k -- the same window size reaches the root on one grammar and stops at the blocks on another, which is the rule-dependence hshunt expected.

Only medians over at least 3 grammars are drawn. The gap at L=3, k=5, m=3 is the one place the method runs out: those window sets never saturated within 6M grids (|W| was still growing past 40M windows), and an unsaturated W rejects grammatical grids too, so those cells are excluded rather than reported.

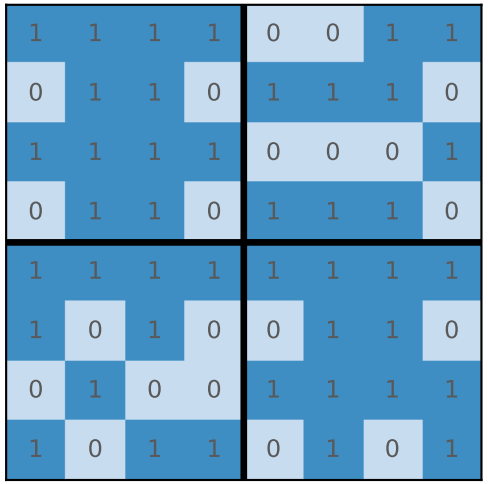
Caveat worth stating plainly: a_j measures the SUPPORT the window set defines -- what an infinite-data machine could reject. It is not what the finite-P sampler actually emits, which is lower; see the sampling-complexity deck.

4. Tokenization, and two worked grids

E_2: four INDEPENDENT legal 4x4 blocks
(coherent to level 2, no common root)
1 of 64 windows rejected



a grammatical grid (E_3)
every window is one the machine knows
0 of 64 windows rejected



Tokenization. A grid is d x d = 8 x 8 SITES, each one token from an alphabet of v symbols; a token is an integer label, with no sub-token structure and no embedding. The root symbol is uniform on {0..v-1} and L levels of rules expand each symbol into an s x s block chosen from m synonymous right-hand sides. The LS machine never sees the tree -- only the leaf grid and the k x k windows cut from it. A window is counted as one int64, its k*k tokens packed base v+1 (the extra digit marks off-grid padding at the border), which is exactly why k >= 7 is out of reach: 3^48 overflows int64.

Left: an E_2 grid for v=2, m=2, L=3, seed 1000. Each 4x4 quadrant is a legal rendering of some level-2 symbol, but the four were drawn independently, so no root produces all four together. The machine rejects 1 of its 64 windows -- boxed in red, centred on the bold site -- and the box straddles the quadrant boundary. That single window is the entire effect: nothing inside any 4x4 block is wrong, and a block-aligned reader sees nothing.

Right: a genuinely grammatical grid from the same model, for contrast -- 0 of 64 rejected, which is the a_L = 1 saturation check made concrete. W here holds 92080 distinct 5x5 windows, grown from 0.7M grammatical grids and matching the exact count N_5 = 92080.

5. Every cell, five rule seeds collapsed

L	v	m	k	j*	sat	med |W|	a_1 (min-max)	a_2 (min-max)	a_L	up
2	2	2	3	1	5/5	216	0.542 (0.261-0.689)	1.000 (1.000-1.000)	1.0000	1
2	2	2	5	1	5/5	818	0.251 (0.250-0.544)	1.000 (1.000-1.000)	1.0000	1
2	2	3	3	1	5/5	402	0.844 (0.472-1.000)	1.000 (1.000-1.000)	1.0000	0-1
2	2	3	5	1	5/5	3330	0.514 (0.314-0.801)	1.000 (1.000-1.000)	1.0000	1
2	2	4	3	1	5/5	512	0.853 (0.674-1.000)	1.000 (1.000-1.000)	1.0000	0-1
2	2	4	5	1	5/5	7540	0.499 (0.313-0.736)	1.000 (1.000-1.000)	1.0000	1
2	2	5	3	1	5/5	646	0.993 (0.826-1.000)	1.000 (1.000-1.000)	1.0000	0-1
2	2	5	5	1	5/5	14174	0.713 (0.522-0.859)	1.000 (1.000-1.000)	1.0000	1
2	2	6	3	1	5/5	742	1.000 (0.956-1.000)	1.000 (1.000-1.000)	1.0000	0-1
2	2	6	5	1	5/5	35316	0.662 (0.539-0.914)	1.000 (1.000-1.000)	1.0000	1
2	3	2	3	1	5/5	454	0.092 (0.077-0.196)	1.000 (1.000-1.000)	1.0000	1
2	3	2	5	1	5/5	1428	0.075 (0.074-0.154)	1.000 (1.000-1.000)	1.0000	1
2	3	3	3	1	5/5	1499	0.230 (0.114-0.370)	1.000 (1.000-1.000)	1.0000	1
2	3	3	5	1	5/5	7524	0.116 (0.098-0.254)	1.000 (1.000-1.000)	1.0000	1
2	3	4	3	1	5/5	3883	0.307 (0.288-0.568)	1.000 (1.000-1.000)	1.0000	1
2	3	4	5	1	5/5	30660	0.148 (0.123-0.276)	1.000 (1.000-1.000)	1.0000	1
2	3	5	3	1	5/5	6387	0.458 (0.385-0.581)	1.000 (1.000-1.000)	1.0000	1
2	3	5	5	1	5/5	65000	0.222 (0.134-0.256)	1.000 (1.000-1.000)	1.0000	1
2	3	6	3	1	5/5	8501	0.604 (0.518-0.828)	1.000 (1.000-1.000)	1.0000	1
2	3	6	5	1	5/5	131060	0.208 (0.173-0.306)	1.000 (1.000-1.000)	1.0000	1
3	2	2	3	1	5/5	222	0.703 (0.392-1.000)	1.000 (0.988-1.000)	1.0000	0-2
3	2	2	5	1	5/5	42387	0.002 (0.001-0.079)	0.648 (0.264-0.860)	1.0000	2
3	2	3	3	1	5/5	396	1.000 (0.336-1.000)	1.000 (1.000-1.000)	1.0000	0-1
3	2	3	5	1	1/5	190891	0.507 (0.507-0.507)	0.965 (0.965-0.965)	1.0000	2
3	3	2	3	1	5/5	1184	0.074 (0.006-0.244)	0.713 (0.510-0.855)	1.0000	2
3	3	2	5	1	5/5	251543	0.000 (0.000-0.000)	0.101 (0.076-0.197)	1.0000	2
3	3	3	3	1	5/5	3904	0.940 (0.171-1.000)	0.999 (0.950-1.000)	1.0000	0-2
3	3	3	5	1	0/5	--	W never saturated --	--	--	
3	4	2	3	1	5/5	3395	0.000 (0.000-0.005)	0.414 (0.384-0.565)	1.0000	2
3	4	2	5	1	5/5	612244	0.000 (0.000-0.000)	0.032 (0.031-0.047)	1.0000	2
3	4	3	3	1	5/5	12562	0.341 (0.294-0.840)	0.899 (0.866-0.980)	1.0000	2
3	4	3	5	1	0/5	--	W never saturated --	--	--	

sat = rule seeds whose window set provably saturated (a_L = 1), out of 5. a_1 and a_2 are medians over those seeds with the full seed range in brackets; up = levels above j* the window detects, as a range over seeds. The 14 cells whose W never saturated (all of them L=3, k=5, m=3) are reported as such rather than dropped silently.